The Rose Operator: Bridging Composition Maps and Transfer Kernels via Local Linear Models

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1 Local Linearizations

Consider the Cauchy problem

$$\frac{du}{dt} + A(u(t)) = 0, \quad t > 0, \tag{1}$$
$$u(0) = u_0, \quad u_0 \in U \subset X,$$

where A is an operator (possibly nonlinear and state-dependent) that generates a contraction semigroup $T(h)$ on the Banach space $\mathcal{X}$.

In many practical settings, one only has access to discrete observations of the state. Let these observations be arranged in the data matrix

$$X_t = \begin{bmatrix} x(t) \\ x(t-h) \\ x(t-2h) \\ \vdots \end{bmatrix}, \tag{2}$$

where the rows correspond to states sampled at time intervals of length h.

A first approach is to attempt a global linearization via a least-squares solution to,

$$X_t \approx X_{t-h}\, T(h), \tag{3}$$
$$\text{or equivalently,} \quad u(t) \approx T(h)\, x(t-h). \tag{4}$$

If the residuals from this approximation are well within an acceptable and prescribed tolerance—then it is reasonable to assume that the dynamics are well represented by a linear operator. In such a case, one can recover the global generator via

$$A = \lim_{h \to 0^+} \frac{1}{h} \left(T(h) - I \right). \tag{5}$$

This approach is particularly effective when A is linear or nearly linear. The contraction property aids in establishing the existence and uniqueness of solutions, however it also means that a global linearization of A may be unattainable when its dependence on $u(t)$ is significant with residuals remaining unacceptably large.

A global linearization may fail to capture the system's behavior when A is state dependent. In such case a local linearization may be preferable.

To construct a local approximation we can introduce a weighting operator W designed to select or emphasize those rows in X_t that are dynamically relevant. The idea is to filter the data so that regions exhibiting dynamics similar to those surrounding $x(t)$ are highlighted, typically by considering observations local to $x(t)$ under some metrization of $\mathcal{X}$.

If W is constructed using a localization kernel, $k_\theta(x(j), x(i))$, where θ parametrizes what "local" means, then the weighted observations are,

$$W_\theta X_t, \tag{6}$$

and the local approximation is expressed as

$$W_\theta X_t \approx W_\theta X_{t-h}\,\hat{T}(h,\theta,t), \tag{7}$$

where $\hat{T}(h,\theta,t)$ is the locally linearized evolution operator valid in a neighborhood of $x(t)$. If a weighted pseudoinverse is employed,

$$A(x(t),\theta,h) = \Big((W_\theta X_{t-h})^\dagger\,(W_\theta X_t) - I \Big) \tag{8}$$

where $(WX_{t-h})^+$ denotes the pseudoinverse of WX_{t-h}. Now recovery of the generator $A(x(t))$ requires the spatial limit in θ, followed by the temporal limit in h,

$$A(x(t)) = \lim_{h\to 0}\frac{1}{h}\left(\lim_{\theta\to 0}\hat{T}(h,\theta,t) - I\right) \tag{9}$$

If T is (i) deterministic (i.e. $\theta \to 0$ iff, in the weak sense, $k_\theta \to \delta$), and (ii) locally linear (i.e. $T = \sum_{k=0}^{\infty}\frac{h^k}{k!}A(h,\theta,x_j)^k$ converges where required), then this is a nice framework for approximating the state-dependent generator $A(x(t))$ in the presence of features precluding a global linear model.

However, in the absence of any strong *a priori* intuition regarding even the most general assumptions of the generating process, most notably stochasticity and the rate of decay of higher order terms, deciding how to approach these limits (and thus what constitutes a *good* linearization) is not easy.

2 Composition and Transfer Operators

Definition 2.1 (Composition (Koopman) operator)**.** Let $T_h : X \to X$ be the time–h map of the flow (1). For any square–integrable observable $f \in L^2(X,\mu)$ define

$$(Kf)(x) = f\big(T_h x\big)$$

- K is linear but acts on an *infinite–dimensional* function space even when $\dim X < \infty$.
- If T_h preserves μ then K is unitary and its spectrum lies on the unit circle.
- No smoothing occurs: higher–order nonlinearities of the flow are implicit in composition, so K captures sharp deterministic features but is blind to stochasticity.
- For the Dirac kernel $p_\delta(x' \mid x) = \delta\big(x' - T_h x\big)$, K is the *left* adjoint of the deterministic Perron–Frobenius operator P_δ (see (10) below).

Global linearisation (3) corresponds to seeking a finite–rank approximation of K.

Definition 2.2 (Transfer (Perron–Frobenius) operator)**.** Let $p(x' \mid x)$ be a Markov transition density. For any $\rho \in L^1(X,\mu)$ set

$$(P\rho)(x') = \int_X \rho(x)\,p(x' \mid x)\,d\mu(x) \tag{10}$$

- P is positive and mass–preserving: $\int (P\rho)\, d\mu = \int \rho\, d\mu$.
- Under mild regularity P is a strict contraction on the mean–zero subspace $L_0^2(X)$ whenever p has non–zero width; its spectral radius is then < 1.
- A non–trivial width encodes diffusion, measurement error, or model misspecification. In local linearisation the residual cloud $r_t^{(\theta)}$ naturally induces such a kernel.
- For the deterministic kernel $p_\delta(x' \mid x) = \delta\big(x' - T_h x\big)$, P_δ is the *right* adjoint of the Koopman operator (2.1).

Throughout we fix a reference measure μ; all L^p spaces, inner products, and adjoints are taken with respect to μ.

Two extreme lenses. Koopman K captures deterministic curvature but ignores noise, whereas a non–delta transfer operator P absorbs all residual variation as diffusion. Global least–squares (3) implicitly demands that K admit an accurate finite–rank approximation; large residuals signal failure. Conversely, shunting all residual energy into P treats it as pure diffusion.

Need for a continuous bridge. We therefore seek a *continuous family* of operators

$$(\theta, \sigma) \;\longmapsto\; \big(\mathcal{R}^{\flat}_{\theta,\sigma}, \mathcal{R}^{\sharp}_{\theta,\sigma}\big)$$

such that $\lim_{\theta\to 0,\, \sigma\to 0} \mathcal{R}^{\flat}_{\theta,\sigma} = K$ and $\lim_{\theta,\sigma\to\infty} \mathcal{R}^{\sharp}_{\theta,\sigma} = P_\infty$ (the maximally diffusive rank-one PF operator). The construction is *local* (uses weighted neighbourhoods) and *data-adaptive* (learns residual covariance), producing a a two-parameter homotopy surface whose θ–axis refines deterministic drift and whose σ–axis injects diffusion.

The next section formalises this via the push-forward / pull-back pair $(\mathcal{R}^{\sharp}_{\theta,\sigma}, \mathcal{R}^{\flat}_{\theta,\sigma})$ introduced in Definition (3.1).

3 From Koopman and PF to a Two–Dial Family

The preceding section set the deterministic pull–back (Koopman) and stochastic push–forward (Perron–Frobenius) operators at opposite ends of the modeling spectrum. We now construct a *two-parameter* family of operators

$$(\theta, \sigma) \;\longmapsto\; \big(\mathcal{R}^{\flat}_{\theta,\sigma}, \mathcal{R}^{\sharp}_{\theta,\sigma}\big), \qquad (\theta,\sigma) \in (0,\infty) \times (0,\infty)^{1} \tag{11}$$

with the following properties:

1. **Koopman corner**: $\displaystyle\lim_{\theta\to 0,\, \sigma\to 0} \mathcal{R}^{\flat}_{\theta,\sigma} = K$ and, dually, $\displaystyle\lim_{\theta\to 0,\, \sigma\to 0} \mathcal{R}^{\sharp}_{\theta,\sigma} = P_\delta$ (the deterministic PF operator with Dirac kernel).

[1] Because $(0,\infty)$ is homeomorphic to $[0,1)$ via $u \mapsto \frac{1}{(1+u)}$ the homotopy can be re-parameterised to the unit interval without affecting continuity; we retain the positive bandwidth scales for engineering clarity.

2. **Maximally diffusive edge**: $\lim_{\theta,\sigma\to\infty} \mathcal{R}^{\sharp}_{\theta,\sigma} = P_\infty$ where P_∞ is the rank-one transfer operator whose kernel is a spatially uniform density, and $\mathcal{R}^{\flat}_{\theta,\sigma}$ collapses observables to the corresponding constant.
3. **Continuity and commutativity**: The map $(\theta,\sigma) \mapsto (\mathcal{R}^{\flat}_{\theta,\sigma}, \mathcal{R}^{\sharp}_{\theta,\sigma})$ is jointly continuous in the L^2-operator norm; varying θ (locality) and σ (residual blur) in either order yields the same result, giving a *homotopy of homotopies*—a filled square in the 2-category $\mathrm{End}(L^2(X,\mu))$.

We call this two-dial construction the **Rose operator family**. Operationally it couples

- a *locally weighted linear predictor* $C_x^{(\theta)}$ that refines deterministic drift as $\theta \downarrow 0$, and
- an *empirical residual kernel* governed by g_σ that injects increasing diffusion as $\sigma \uparrow \infty$.

Because the θ- and σ-axes commute, one may independently *sharpen drift* (decrease θ) or *blur noise* (increase σ), tracing paths across the square that suit the data at hand. In the interior of this "Rose surface" lie operator pairs that blend deterministic curvature with state– dependent or homogeneous diffusion—precisely the regime where, presumably, most representations arise.

3.1 Two–parameter Rose operator $\mathcal{R}_{\theta,\sigma}$

Locality kernel ϕ (state space). Fix a radial profile $\phi : [0,\infty) \to \mathbb{R}_+$ satisfying

$$\int_{\mathbb{R}^d} \phi(\|u\|)\, du = 1 \qquad \phi(r) = \phi(-r) \qquad \int r^2\phi(r)\, dr < \infty \qquad \|\phi_{\lambda_1} - \phi_{\lambda_2}\|_{L^1} \le L\,|\lambda_1 - \lambda_2| \quad (12)$$

with $\phi_\lambda(r) := \lambda^{-d}\phi(r/\lambda)$. Typical choices include Gaussian, Epanechnikov, tri-weight, or a heat kernel on a Riemannian manifold. For bandwidth $\theta > 0$ define the neighbourhood weight

$$w_\theta\big(\|x_t - x\|\big) = \phi_\theta\big(\|x_t - x\|\big) \tag{13}$$

Local drift and residuals. With $W_\theta = \mathrm{diag}\big(w_\theta(\|x_t - x\|)\big)$ and $X = [x_0, \ldots, x_{N-1}]$, $Y = [x_h, \ldots, x_{N-1+h}]$, set

$$C_x^{(\theta)} = Y_\theta X_\theta^\top \left(X_\theta X_\theta^\top\right)^\dagger, \qquad r_t^{(\theta)}(x) = x_{t+h} - C_x^{(\theta)} x_t \tag{14}$$

Residual kernel g_σ (noise space). Let $g_\sigma : \mathbb{R}^d \to \mathbb{R}_+$ be radial with $\int g_\sigma = 1$ and $g_\sigma \to \delta$ as $\sigma \downarrow 0$ (e.g. Gaussian, Huber, Tukey). Define first

$$Z_{\theta,\sigma}(x) = \sum_t w_\theta(\|x_t - x\|)\, g_\sigma\big(r_t^{(\theta)}(x)\big) \tag{15}$$

and then the weighted residual mean

$$\bar{r}_x^{(\theta,\sigma)} = \frac{1}{Z_{\theta,\sigma}(x)} \sum_t w_\theta(\|x_t - x\|)\, g_\sigma\big(r_t^{(\theta)}\big)\, r_t^{(\theta)} \tag{16}$$

as well as the covariance

$$\Sigma_x^{(\theta,\sigma)} = \frac{1}{Z_{\theta,\sigma}(x)} \sum_t w_\theta(\|x_t - x\|)\, g_\sigma\big(r_t^{(\theta)}\big)\, \big(r_t^{(\theta)} - \bar{r}_x^{(\theta,\sigma)}\big)\big(r_t^{(\theta)} - \bar{r}_x^{(\theta,\sigma)}\big)^\top \tag{17}$$

Transition kernel.

$$k_{\theta,\sigma}(y \mid x) = \mathcal{N}\big(y;\ C_x^{(\theta)} x + \bar{r}_x^{(\theta,\sigma)}, \Sigma_x^{(\theta,\sigma)}\big) \tag{18}$$

Composition and Transfer Adjunctions.

Definition 3.1 (Push–forward and pull-back Rose operators)**.** For the kernel $k_{\theta,\sigma}$ defined above set,

$$(\mathcal{R}^{\sharp}_{\theta,\sigma}\rho)(y) := \int_X k_{\theta,\sigma}(y \mid x)\, \rho(x)\, d\mu(x), \qquad (\mathcal{R}^{\flat}_{\theta,\sigma} f)(x) := \int_X f(y)\, k_{\theta,\sigma}(y \mid x)\, d\mu(y) \tag{19}$$

Then $\mathcal{R}^{\flat}_{\theta,\sigma}$ is the Hilbert-space adjoint of $\mathcal{R}^{\sharp}_{\theta,\sigma}$ with respect to the $L^2(X,\mu)$ pairing,

$$\langle f, \mathcal{R}^{\sharp}_{\theta,\sigma}\rho\rangle_{L^2} = \langle \mathcal{R}^{\flat}_{\theta,\sigma} f, \rho\rangle_{L^2} \qquad \text{for all } f, \rho \in L^2(X,\mu) \tag{20}$$

3.2 Homotopy theorems for the two–dial Rose family

The results below make precise the idea that the two–parameter Rose operators

$$(\theta,\sigma) \mapsto (\mathcal{R}^{\flat}_{\theta,\sigma}, \mathcal{R}^{\sharp}_{\theta,\sigma}) \tag{21}$$

form a *continuous square* whose left–hand edge collapses to the Koopman / deterministic–PF pair and whose right–hand edge collapses to the rank-one, maximally diffusive PF pair.

Assumption 3.1 (Standing hypotheses)*.*

1. $X \subset \mathbb{R}^d$ is a compact C^∞ manifold equipped with a finite Borel measure μ.
2. $T_h : X \to X$ is the time-h map of a C^2 flow with bounded first and second derivatives, $\sup_x \|DT_h(x)\| \le M_1$, $\sup_x \|D^2 T_h(x)\| \le M_2$.
3. The locality profile ϕ is radial, integrable, symmetric, finite-moment, and Lipschitz in scale (Section 3.1).
4. The residual-weight profile g_σ is radial with $\int g_\sigma = 1$ and $g_\sigma \to \delta$ in L^1 as $\sigma \downarrow 0$.

Theorem 3.1 (Two–parameter Rose homotopy)**.** *Under Assumption 3.1, in the population (infinite–data) setting, the following statements hold.*

(i) ***Local bias and covariance bounds***

For every $x \in X$ and $\sigma > 0$

$$\|C_x^{(\theta)} - DT_h(x)\| \le c_1\,\theta, \qquad \|\Sigma_x^{(\theta,\sigma)}\| \le c_2\,(\theta^2 + \sigma^2)^2 \tag{22}$$

(ii) ***Deterministic corner***

$$\lim_{\theta\to 0,\ \sigma\to 0} \|\mathcal{R}^{\flat}_{\theta,\sigma} - K\|_{L^2\to L^2} = 0, \qquad \lim_{\theta\to 0,\ \sigma\to 0} \|\mathcal{R}^{\sharp}_{\theta,\sigma} - P_\delta\|_{L^2\to L^2} = 0 \tag{23}$$

(iii) ***Maximally diffusive edge***

$$\lim_{\theta,\sigma\to\infty} \|\mathcal{R}^{\sharp}_{\theta,\sigma} - P_\infty\|_{L^2\to L^2} = 0, \qquad P_\infty\rho = \Big(\int_X \rho\, d\mu\Big) p_\infty \tag{24}$$

where p_∞ is the state–independent limit density.

(iv) ***Separate Lipschitz continuity***

There exist constants $C_\theta, C_\sigma > 0$ such that

$$\|\mathcal{R}^{\flat}_{\theta_1,\sigma} - \mathcal{R}^{\flat}_{\theta_2,\sigma}\|_{L^2\to L^2} \le C_\theta\, |\theta_1 - \theta_2|, \qquad \|\mathcal{R}^{\flat}_{\theta,\sigma_1} - \mathcal{R}^{\flat}_{\theta,\sigma_2}\|_{L^2\to L^2} \le C_\sigma\, \|g_{\sigma_1} - g_{\sigma_2}\|_{L^1} \tag{25}$$

and the same bounds hold for the push–forward operators $\mathcal{R}^\sharp$.

(v) ***Commutativity (2-cell)***

For any $\theta_2 \ge \theta_1 > 0$ and $\sigma_2, \sigma_1 > 0$,

$$\mathcal{R}_{\theta_2,\sigma_2} = \mathcal{R}_{\theta_2,\sigma_2-\sigma_1} \circ \mathcal{R}_{\theta_1,\sigma_1} = \mathcal{R}_{\theta_1,\sigma_1} \circ \mathcal{R}_{\theta_2-\theta_1,\sigma_2} \tag{26}$$

whenever g_σ is Gaussian; more generally the two one-parameter homotopies commute up to a vanishing L^2-operator error of order $O\big(|\theta_2 - \theta_1|\, \|g_{\sigma_2} - g_{\sigma_1}\|_{L^1}\big)$.

Constants $c_1, c_2, C_\theta, C_\sigma$ depend only on M_1, M_2, the second moment of ϕ, and integrability bounds for g.

Sketch of proof. (i) A second-order Taylor expansion plus symmetry of ϕ yields the bias bound; the covariance bound follows by inserting $g_\sigma \le \|g_\sigma\|_\infty$ and taking second moments.

(ii) As $\theta, \sigma \downarrow 0$, $C_x^{(\theta)} \to DT_h(x)$ and $\Sigma_x^{(\theta,\sigma)} \to 0$; the Gaussian TV estimate gives $k_{\theta,\sigma} \to \delta_{T_h x}$ in L^1, hence operator convergence.

(iii) With $\theta, \sigma \uparrow \infty$, locality weights flatten and g_σ acts as heavy blur: the kernel columns converge to the state–independent density p_∞.

(iv) Lipschitz in θ is inherited from Lipschitz scale dependence of ϕ; Lipschitz in σ follows from $\|g_{\sigma_1} - g_{\sigma_2}\|_{L^1}$.

(v) For Gaussian g_σ, convolution is additive in variances, so ordering of the two dials commutes exactly; for general radial kernels one obtains the stated error estimate by Young's inequality. □

4 Continuous–Time Limit and Stochastic Interpretation

The discrete–time Rose operators $(\mathcal{R}^{\flat}_{\theta,\sigma}, \mathcal{R}^{\sharp}_{\theta,\sigma})$ approximate, in the small–lag limit, the semigroup of an Itô diffusion. This section formalises that link and shows how the local bandwidth–weighted quantities $C_x^{(\theta)}$ and $\Sigma_x^{(\theta,\sigma)}$ furnish non-parametric estimators of the drift $a(x)$ and diffusion tensor $D(x)$ of the underlying continuous dynamics.

Throughout this section $h > 0$ denotes the sampling interval and we explicitly write $C_x^{(\theta,h)}$, $r_t^{(\theta,h)}$, $\Sigma_x^{(\theta,\sigma,h)}$ to highlight the dependence on h. Assumptions 3.1 remain in force.

Lemma 4.1 (First-order expansion)**.** *For every fixed $\theta > 0, \sigma > 0$ and anchor $x \in X$*

$$C_x^{(\theta,h)} = I + a(x)\,h + O\big(h\,\theta + h^2\big), \qquad \Sigma_x^{(\theta,\sigma,h)} = 2D(x)\,h + O\big(h\,\theta^2 + h\,\sigma^2\big) \tag{27}$$

where $a(x) = \dot{X}_t|_{t=0}$ is the deterministic drift and $D(x) = \frac{1}{2}B(x)B(x)^\top$ is the diffusion tensor of the Itô SDE $dX_t = a(X_t)\,dt + B(X_t)\,dW_t$.

Sketch. Second-order Taylor expansion of T_h yields $T_h x_t = x + a(x)h + O(h\,\|x_t - x\| + h^2)$. Weighted first-moment symmetry of ϕ eliminates the linear $x_t - x$ term, giving the stated drift expansion. For the covariance, subtract $\bar{r}_x^{(\theta,\sigma,h)}$ and take second moments; the leading term is $2D(x)h$ and the weights introduce the stated $O(h\theta^2 + h\sigma^2)$ correction. □

Theorem 4.1 (Discrete–to–continuous limit)**.** *Fix $\theta, \sigma > 0$ and let $h \downarrow 0$. For any $f \in C^2(X)$ and $\rho \in C^2(X)$,*

$$\frac{\mathcal{R}^{\flat,h}_{\theta,\sigma} f - f}{h} \longrightarrow \mathcal{L}^\flat f, \qquad \frac{\mathcal{R}^{\sharp,h}_{\theta,\sigma} \rho - \rho}{h} \longrightarrow \mathcal{L}^\sharp \rho \quad \text{in } L^2(X,\mu) \tag{28}$$

where the Kolmogorov generators are

$$\mathcal{L}^\flat f = a\cdot \nabla f + D : \nabla\nabla f, \qquad \mathcal{L}^\sharp \rho = -\nabla\cdot (a\rho) + \nabla\cdot (D\nabla\rho) \tag{29}$$

corresponding to the Itô SDE above.

Sketch. Expand $f(y)$ to second order around x and integrate against the Gaussian kernel with mean $x + a(x)h$ and variance $2D(x)h$; the first-order terms reproduce $\mathcal{L}^\flat f$. An identical calculation with adjoint pairing yields the forward generator $\mathcal{L}^\sharp$. □

Remark 4.1. Lemma 4.1 yields *data-driven, bandwidth-controlled* estimators for both coefficients of the limiting SDE, namely $\widehat{a}(x) = (C_x^{(\theta,h)} - I)/h$ for the drift and $\widehat{D}(x) = \frac{1}{2}\,\Sigma_x^{(\theta,\sigma,h)}/h$ for the diffusion tensor. Once these fields are available, the generator pair $(\mathcal{L}^\flat, \mathcal{L}^\sharp)$ provides two complementary routes to uncertainty propagation: forward evolution of densities through the Fokker–Planck equation and simulation of sample paths via the Itô SDE, with the discrete Rose operators furnishing time-h stepping schemes that are consistent with both perspectives.

5 Fitting Algorithm

Consider observations of state transition pairs $(y_i, x_i)_{i\in N}$ where $y_i = x_{i+1}$.

Algorithm 5.1 (H). Data-Driven Delay Embedding and Pointwise θ Selection via Rose+PCA

Require: Univariate series $\{x(t)\}$, over-embed dimension D, grid size K, variance threshold $\alpha \in (0,1)$.

Ensure: Reduced dimension d, per-anchor $\{\theta_i^*\}$, final residuals $\{\widetilde{r}_i \in \mathbb{R}^d\}$, coefficients $\{\widetilde{C}_i \in \mathbb{R}^{d\times d}\}$.

1. Build full D-dimensional delay embedding:
for $i = 1$ to N **do**
  $\mathbf{x}_i \leftarrow \big[x(t_i),\, x(t_i - 1),\, x(t_i - 2),\, \ldots,\, x(t_i - (D-1))\big] \in \mathbb{R}^D$.
end for
$E \leftarrow [\mathbf{x}_1; \ldots; \mathbf{x}_N] \in \mathbb{R}^{N\times D}$.

2. Construct θ-grid
$\theta \leftarrow [\theta_0 = 0; \ldots; \theta_K = \theta_{max}]$
4. Compute Rose residuals $r_{i,k} \in \mathbb{R}^D$:
for $i = 1$ to N **do**
  for $k = 1$ to K **do**
    Fit WLS at $\mathbf{x}_i$ with kernel scale θ_k, forecast next step $\widehat{\mathbf{x}}_i$.
    $r_{i,k} \leftarrow \mathbf{x}(t_i + 1) - \widehat{\mathbf{x}}_i$.
  end for
end for
5. Select pointwise θ_i^*:
for $i = 1$ to N **do**
  $\rho_i(k) \leftarrow \|r_{i,k}\|_2, \quad k_i^* \leftarrow \arg\min_{1 \le k \le K} \rho_i(k)$.
  $\theta_i^* \leftarrow \theta_{k_i^*}$.
end for
6. Collect selected residuals in $\mathbb{R}^D$:
for $i = 1$ to N **do**
  $r_i^* \leftarrow r_{\,i,\,k_i^*}$.
end for
$R^* \leftarrow [\, r_1^*; \, \ldots; \, r_N^*] \in \mathbb{R}^{N \times D}$.
7. PCA on R^* to determine reduced dimension d:
Compute SVD: $R^* = U \, \Sigma \, V^\top$.
Choose smallest d with $\sum_{j=1}^{d} \sigma_j^2 \, / \, \sum_{j=1}^{D} \sigma_j^2 \ge \alpha$.
$V_d \leftarrow V[:, 1:d] \in \mathbb{R}^{D \times d}$.
8. Form reduced embedding in $\mathbb{R}^d$:
for $i = 1$ to N **do**
  $\widetilde{\mathbf{x}}_i \leftarrow V_d^\top \, \mathbf{x}_i \in \mathbb{R}^d$.
end for
9. Re-fit Rose in $\mathbb{R}^d$ at θ_i^*:
for $i = 1$ to N **do**
  Fit WLS on $\widetilde{\mathbf{x}}_i$ with scale θ_i^*, forecast $\widehat{\widetilde{\mathbf{x}}}_i$.
  $\widetilde{r}_i \leftarrow \widetilde{x}(t_i + 1) - \widehat{\widetilde{\mathbf{x}}}_i \in \mathbb{R}^d$.
  Extract coefficient $\widetilde{C}_i \in \mathbb{R}^{d \times d}$.
end for
10. Save outputs via save_all**:**
$\{\widetilde{r}_i\}_{i=1}^{N}, \; \{\widetilde{C}_i\}_{i=1}^{N}, \; \{t_i\}_{i=1}^{N}, \; \{\theta_i^*\}_{i=1}^{N}$.

References

A Notation & Conventions

PX	probability measures on $(X, \mathcal{F})$
G	Giry monad (B.15)
$x \mapsto \delta_x$	Dirac embedding
$\rightsquigarrow$	kernel arrow (Def. B.13)
$\flat, \sharp$	musical conversions (Def. **??**)
P_t, L	Markov semigroup & generator (B.23–B.24)
k_Δ, T	LWLS kernel & Rose operator (**??**–**??**)
Δ	discrete time step in data

B Definitions

B.1 Measure & Probability Foundations

Definition B.1 (σ–Algebra)**.** Given a set X, a *σ–algebra* $\mathcal{F} \subseteq 2^X$ is a non-empty collection of subsets closed under complementation and countable unions (hence also countable intersections).

Definition B.2 (Measurable Space)**.** A *measurable space* is a pair $(X, \mathcal{F})$ where $\mathcal{F}$ is a σ–algebra on X (Definition B.1).

Definition B.3 (Measurable Map)**.** If $(X, \mathcal{F})$ and $(Y, \mathcal{G})$ are measurable spaces, a map $f : X \to Y$ is *measurable* if $f^{-1}(B) \in \mathcal{F}$ for all $B \in \mathcal{G}$.

Definition B.4 (Measure & Probability Measure)**.** Let $(X, \mathcal{F})$ be a measurable space. A *measure* is a countably additive map $\mu : \mathcal{F} \to [0, \infty]$ with $\mu(\varnothing) = 0$. It is a *probability measure* if $\mu(X) = 1$. We write PX for the set of all probability measures on $(X, \mathcal{F})$.

Definition B.5 (Push-Forward Measure)**.** Given a measurable $f : (X, \mathcal{F}) \to (Y, \mathcal{G})$ and $\mu \in PX$, the *push-forward* measure $f_*\mu \in PY$ is defined by $(f_*\mu)(B) = \mu\big(f^{-1}(B)\big)$ for $B \in \mathcal{G}$.

Definition B.6 (Pull-Back Measure)**.** Given a measurable $f : (X, \mathcal{F}) \to (Y, \mathcal{G})$ and $\mu \in PX$, the *pull-back* measure $f^*\mu \in PY$ is defined by $f^*\mu(A) \ := \ \mu\big(f(A)\big)$ for $A \in \mathcal{F}$.

B.2 Category Theory

Definition B.7 (Category)**.** A *category* $\mathcal{C}$ consists of a class of objects, hom-sets $\mathrm{Hom}_{\mathcal{C}}(A, B)$ for each ordered pair (A, B), identity morphisms id_A, and associative composition, satisfying the usual unit laws.

Definition B.8 (Functor)**.** Let $\mathcal{C}, \mathcal{D}$ be categories. A *functor* $F : \mathcal{C} \to \mathcal{D}$ assigns to each object A of $\mathcal{C}$ an object FA of $\mathcal{D}$ and to each morphism $f : A \to B$ a morphism $Ff : FA \to FB$, preserving identities and composition.

Definition B.9 (Finite products in a category)**.** Let $\mathcal{C}$ be a category and $\{ X_i \}_{i=1}^n$ a finite family of objects in $\mathcal{C}$ $(n \geq 0)$.

- A *product* of the family is a pair

$$\Big(X,\ \{\pi_i\}_{i=1}^n\Big), \quad \pi_i : X \longrightarrow X_i \ \text{ in } \mathcal{C},$$

  such that for every object Y and every collection of morphisms $\{\, f_i : Y \to X_i \,\}_{i=1}^n$ there exists a *unique* morphism

$$f : Y \longrightarrow X \qquad \text{with} \qquad \pi_i \circ f = f_i \quad (i = 1, \dots, n).$$

- When it exists, the product is unique up to a unique isomorphism; we write $X = \prod_{i=1}^n X_i$ and call the π_i the *projection morphisms.*
- For $n = 0$ the empty product is any *terminal object* $1 \in \mathcal{C}$ (unique up to unique isomorphism).

$$\begin{tikzcd}
 & Y \arrow[dl, "f_1"'] \arrow[d, dashed, "f"] \arrow[dr, "f_2"] & \\
X_1 & X_1 \times X_2 \arrow[l, "\pi_1"'] \arrow[r, "\pi_2"] & X_2
\end{tikzcd}$$

The dashed arrow f is the *unique* morphism rendering all triangles commutative: $\pi_i \circ f = f_i$ for $i = 1, \dots, n$. This diagram visualises the universal property that characterises the product object $X = \prod_{i=1}^n X_i$.

Definition B.10 (Natural Transformation)**.** Given functors $F, G : \mathcal{C} \to \mathcal{D}$, a *natural transformation* $\eta : F \Rightarrow G$ assigns to every object A a morphism $\eta_A : FA \to GA$ such that $Gf \circ \eta_A = \eta_B \circ Ff$ for all $f : A \to B$ in $\mathcal{C}$.

Definition B.11 (Monad)**.** A *monad* (T, η, μ) on a category $\mathcal{C}$ comprises an endofunctor $T : \mathcal{C} \to \mathcal{C}$, a unit natural transformation $\eta : \mathrm{id}_{\mathcal{C}} \Rightarrow T$ and a multiplication $\mu : T^2 \Rightarrow T$ satisfying $\mu \circ T\eta = \mu \circ \eta T = \mathrm{id}_T$ and $\mu \circ T\mu = \mu \circ \mu T$.

Definition B.12 (Category **Meas**)**.** Objects are measurable spaces; morphisms are measurable maps (Definition B.3).

Definition B.13 (Markov Kernel)**.** Let $(X, \mathcal{F})$ and $(Y, \mathcal{G})$ be measurable spaces. A *Markov kernel* $\kappa : X \rightsquigarrow Y$ is a map $\kappa : X \times \mathcal{G} \to [0, 1]$ such that

1. for each $x \in X$, $B \mapsto \kappa(B \mid x)$ is a probability measure on $(Y, \mathcal{G})$;
2. for each $B \in \mathcal{G}$, $x \mapsto \kappa(B \mid x)$ is $\mathcal{F}$-measurable.

Definition B.14 (Category **Stoch**)**.** Objects are measurable spaces. Morphisms are Markov kernels (Definition B.13). Composition is *Chapman–Kolmogorov* integration:

$$(\kappa * \lambda)(C \,|\, x) = \int_Y \kappa(C \,|\, y)\, \lambda(dy \,|\, x).$$

Definition B.15 (Giry Monad)**.** Define $G : \mathbf{Meas} \to \mathbf{Meas}$ by $G(X, \mathcal{F}) = (PX, \mathcal{P}\mathcal{F})$, where $\mathcal{P}\mathcal{F}$ is the weak σ–algebra generated by $\mu \mapsto \mu(B)$ $(B \in \mathcal{F})$. For $f : (X, \mathcal{F}) \to (Y, \mathcal{G})$ set $Gf = f_*$

(Definition B.5). The *unit* $\eta_X : X \to GX$ is $x \mapsto \delta_x$ and the *multiplication* $\mu_X : GGX \to GX$ is $\mu_X(\Phi)(B) = \int_{PX} \nu(B)\, d\Phi(\nu)$. With these, (G, η, μ) is a monad on **Meas** (Definition B.11); its Kleisli category is **Stoch** (Definition B.14).

Definition B.16 (Stochastic Dynamical System)**.** A *stochastic dynamical system* is a G-*coalgebra*: a measurable map $c : X \to GX$ (Definition B.15). Iterating c via Kleisli composition yields a family of kernels $c^{(n)} : X \rightsquigarrow X$; see Definition B.17.

B.3 Markov Kernels & Stochastic Processes

Definition B.17 (Transition Kernel Family)**.** A family $\{\kappa_t\}_{t\geq 0}$ of kernels $\kappa_t : X \rightsquigarrow X$ is a *transition kernel family* if

$$\kappa_{t+s} = \kappa_t * \kappa_s, \qquad \kappa_0(x, \cdot) = \delta_x.$$

Definition B.18 (Discrete-Time Markov Chain)**.** A *discrete-time Markov chain* on state space X is a pair (κ, μ_0) where $\kappa : X \rightsquigarrow X$ is a kernel and $\mu_0 \in PX$ an initial distribution. The law of X_{n+1} given $X_n = x$ is $\kappa(\cdot \mid x)$.

B.4 Semigroup & Generator Theory

Definition B.19 (Bounded Linear Operators)**.** For a Banach space B, the set $\mathcal{L}(B)$ consists of all linear maps $A : B \to B$ with finite operator norm $||A||_{\mathcal{L}(B)} := \sup_{f\neq 0} \frac{||Af||_B}{||f||_B}$

Equipped with this norm, is itself a Banach space.

Definition B.20 (Operator Semigroup)**.** Let B be a Banach space. A family $\{P_t\}_{t\geq 0} \subset \mathcal{L}(B)$ is called an *operator semigroup* (on B) if

$$P_0 = \mathrm{id}_B, \qquad P_{t+s} = P_t\, P_s \quad \text{for all } s, t \geq 0.$$

Definition B.21 (Uniformly Continuous Semigroup)**.** A family $\{P_t\}_{t\geq 0} \subset \mathcal{L}(B)$ is a *uniformly continuous semigroup* if

$$\lim_{t\downarrow 0} \|P_t - \mathrm{id}\|_{\mathcal{L}(B)} = 0, \qquad P_0 = \mathrm{id}, \qquad \text{for all } t \geq 0$$

Equivalently, $P_t = e^{\,tA}$ for a *bounded* operator $A \in \mathcal{L}(B)$ called the (bounded) generator, and the map $t \mapsto P_t$ is continuous in operator norm.

Definition B.22 (Strongly Continuous Semigroup)**.** A family $\{P_t\}_{t\geq 0} \subset \mathcal{L}(B)$ is a *strongly continuous semigroup* (C_0–semigroup) if

$$\lim_{t\downarrow 0} \|P_t f - f\|_B = 0, \qquad \text{for all } f \in B,\, t \geq 0$$

Definition B.23 (Markov Semigroup)**.** A C_0–semigroup $\{P_t\}_{t\geq 0}$ on $L^p(X, \lambda)$ is a *Markov semigroup* if each P_t is positive and $P_t 1 = 1$.

Definition B.24 (Infinitesimal Generator)**.** For a C_0–semigroup $\{P_t\}_{t\geq 0}$ the *generator* is the (possibly unbounded) operator

$$Lf = \lim_{t\downarrow 0} \frac{P_t f - f}{t},$$

with domain $D(L) = \{f : \text{limit exists in } \|\cdot\|\}$.

C Locality in Infinitesimal Generators

C.1 The Evolution Kernel

In the classical theory of C_0 semigroups, the evolution operator $T(h)$ composes a C_0 semigroup generated by a linear operator A

$$T(h) = e^{hA}.$$

In this setting, the kernel $K_h(\xi, \eta)$ is the Dirac delta

$$K_h(\xi, \eta) \xrightarrow[h\to 0]{} \delta(\xi - \eta).$$

Proof. Consider the integral operator

$$(T(h)f)(\xi) = \int_\Omega K_h(\xi, \eta) f(\eta) d\eta \tag{30}$$

with infinitesimal generator

$$\begin{aligned}
(Af)(\xi) &= \lim_{h\to 0} \frac{1}{h} \left(T(h) - I\right) f(\xi) &(31)\\
&= \lim_{h\to 0} \frac{1}{h} \left((T(h)f)(\xi) - f(\xi)\right) &(32)
\end{aligned}$$

By substitution

$$(Af)(\xi) = \lim_{h\to 0} \frac{1}{h} \left(\int_\Omega K_h(\xi, \eta) f(\eta) d\eta - f(\xi) \right) \tag{33}$$

The goal is to rewrite the integral term so that $f(\xi)$ appears inside the integral. We add and subtract $K_h(\xi, \eta) f(\xi)$ inside the integral,

$$(Af)(\xi) = \lim_{h\to 0} \frac{1}{h} \left(\int_\Omega K_h(\xi, \eta) f(\eta) - K_h(\xi, \eta) f(\xi) + K_h(\xi, \eta) f(\xi) d\eta - f(\xi) \right) \tag{34}$$

and factor,

$$\begin{aligned}
(Af)(\xi) &= \lim_{h\to 0} \frac{1}{h} \left(\int_\Omega K_h(\xi, \eta)(f(\eta) - f(\xi)) d\eta + \int_\Omega K_h(\xi, \eta) f(\xi) d\eta - f(\xi) \right) &(35)\\
&= \lim_{h\to 0} \frac{1}{h} \int_\Omega K_h(\xi, \eta)(f(\eta) - f(\xi)) d\eta + \lim_{h\to 0} \frac{1}{h} \left(\int_\Omega K_h(\xi, \eta) d\eta - 1 \right) f(\xi) &(36)
\end{aligned}$$

We can ensure that $K_h(\xi, \eta)$ acts as a state transition probability density by imposing the normalization condition,

$$\int_\Omega K_h(\xi, \eta) d\eta = 1 \tag{37}$$

while at the same time eliminating the second term,

$$\lim_{h \to 0} \frac{1}{h} \left(\int_\Omega K_h(\xi, \eta) d\eta - 1 \right) f(\xi) = 0 \tag{38}$$

thus recovering the infinitesimal generator as a standard non-local operator,

$$(Af)(\xi) = \lim_{h \to 0} \frac{1}{h} \int_\Omega K_h(\xi, \eta)(f(\eta) - f(\xi)) d\eta \tag{39}$$

Kernel Structure: The resolvent operator associated with A is given by:

$$R_\lambda(A) = (\lambda I - A)^{-1}. \tag{40}$$

It is well known (see, e.g. Pazy) that the semigroup $T(h)$ can be expressed in terms of the resolvent via the contour integral,

$$T(h) = \frac{1}{2\pi i} \oint_\Gamma e^{h\lambda} R_\lambda(A) d\lambda \tag{41}$$

We equate the integral representations

$$\int_\Omega K_h(\xi, \eta) f(\eta) d\eta = \frac{1}{2\pi i} \oint_\Gamma e^{h\lambda} R_\lambda(A) f(\xi) d\lambda \tag{42}$$

Furthermore, Arendt & Bukhvalov showed that if $T(h)$ is an integral operator, then so is the resolvent,

$$(R_\lambda f)(\xi) = \int_\Omega M_\lambda(\xi, \eta) f(\eta) d\eta \tag{43}$$

where $M_\lambda(\xi, \eta)$ is the resolvent kernel. From (42) using the integral resolvent,

$$\int_\Omega K_h(\xi, \eta) f(\eta) d\eta = \frac{1}{2\pi i} \oint_\Gamma e^{h\lambda} \int_\Omega M_\lambda(\xi, \eta) f(\eta) d\eta \, d\lambda \tag{44}$$

Two cases follow regarding the absolute integrability of

$$F(\lambda, \eta) = e^{h\lambda} M_\lambda(\xi, \eta) f(\eta)$$

If $F \in L^1(\Gamma \times \Omega)$ then for

$$\oint_\Gamma \int_\Omega |F(\lambda, \eta)| \, d\mu(\eta) \, d\nu(\lambda) \tag{45}$$

we may apply Fubini's theorem and interchange the order of integration

$$\int_\Omega K_h(\xi,\eta) f(\eta) d\eta = \int_\Omega \left(\frac{1}{2\pi i} \oint_\Gamma e^{h\lambda} M_\lambda(\xi,\eta) d\lambda \right) f(\eta) d\eta \tag{46}$$

Since this must hold for all f, we identify the kernels,

$$K_h(\xi,\eta) = \frac{1}{2\pi i} \oint_\Gamma e^{h\lambda} M_\lambda(\xi,\eta) d\lambda \tag{47}$$

Since $T(0) = I$, the kernel at $h = 0$ must satisfy

$$K_0(\xi,\eta) = \delta(\xi - \eta) \tag{48}$$

In many classical settings, one can show that the resolvent satisfies

$$(\lambda I - A) R_\lambda(A) = I \tag{49}$$

so that in kernel form

$$(\lambda I - A) M_\lambda(\xi,\eta) = \delta(\xi - \eta) \tag{50}$$

$M_\lambda(\xi,\eta)$ serves as the Green's function for the operator $\lambda I - A$. Supposing that for values of λ in a suitable region the Neumann series expansion hold

$$(\lambda I - A)^{-1} = \sum_{n=0}^{\infty} \frac{A^n}{\lambda^{n+1}} \tag{51}$$

Then the kernel $M_\lambda(\xi,\eta)$ may be expressed as

$$\tag{52}$$

$$M_\lambda(\xi,\eta) = \sum_{n=0}^{\infty} \frac{A^n}{\lambda^{n+1}} \delta(\xi - \eta) \tag{53}$$

Substitute this expansion into

$$K_h(\xi,\eta) = \frac{1}{2\pi i} \oint_\Gamma e^{h\lambda} \left(\sum_{n=0}^{\infty} \frac{A^n}{\lambda^{n+1}} \delta(\xi - \eta) \right) d\nu(\lambda) \tag{54}$$

$$= \delta(\xi - \eta) \sum_{n=0}^{\infty} A^n \left[\frac{1}{2\pi i} \oint_\Gamma \frac{e^{h\lambda}}{\lambda^{n+1}} d\nu(\lambda) \right] \tag{55}$$

According to the residue theorem

$$\frac{1}{2\pi i} \oint_\Gamma \frac{e^{h\lambda}}{\lambda^{n+1}} d\nu(\lambda) = \frac{h^n}{n!} \tag{56}$$

Substituting this back into (55), we obtain

$$K_h(\xi,\eta) = \delta(\xi - \eta) \sum_{n=0}^{\infty} \frac{h^n}{n!} A^n = \delta(\xi - \eta) e^{hA} \tag{57}$$

Thus,

$$K_h(\xi,\eta) = \delta(\xi-\eta)e^{hA} \tag{58}$$

which is equivalent to the operator identity,

$$T(h) = e^{hA} \tag{59}$$

□

The Nonlinear (State-Dependent) Case: When the system is nonlinear, the evolution is typically given by a state-dependent mapping,

$$T(h)(\xi) = \Phi_h(\xi),$$

we may still maintain an integral representation:

$$(T(h)f)(\xi) = \int_\Omega K_h(\xi,\eta)f(\eta)\,d\eta.$$

In order for the kernel to capture the correct local behavior of the flow, we require that for small h,

$$K_h(\xi,\eta) \to \delta\Big(\Phi_h(\xi) - \Phi_h(\eta)\Big).$$

Since Φ_0 is the identity mapping, this condition reduces to

$$\lim_{h\to 0} K_h(\xi,\eta) = \delta(\xi-\eta),$$

which, in the weak sense, means that for any test function f,

$$\lim_{h\to 0}\int_\Omega K_h(\xi,\eta)f(\eta)\,d\eta = f(\xi).$$

D A Categorical Scaffold for Local Linearization

Local linear models, residual covariances, and adaptive basis updates can all be written algebraically, but a categorical formulation clarifies *which parts belong to the dynamics, which to the coordinate representation, and which to the data–driven closure.* By lifting the construction to functors, natural transformations, and (co)monads we obtain

(i) a precise separation of concerns

(ii) a route to operator-theoretic results via functoriality

(iii) and a universal property (*cofree* coalgebra) that characterises the residual-frame augmentation

D.1 Components

Throughout, $k \geq 2$ is fixed, so that C^k flows have well-defined first and second jets. All manifolds are assumed compact, and all vector spaces are over $\mathbb{R}$.

Definition D.1 (**Dyn**)**.** The category **Dyn** consists of

- **objects** $(\mathcal{M}, \varphi^t)$ where $\mathcal{M}$ is a compact C^k manifold and $\varphi^t : \mathcal{M} \to \mathcal{M}$ a C^k flow.
- **morphisms** $\Psi : (\mathcal{M}, \varphi^t) \to (\mathcal{N}, \psi^t)$ given by C^k maps $\Psi : \mathcal{M} \to \mathcal{N}$ satisfying
$$\Psi \circ \varphi^t = \psi^t \circ \Psi \tag{60}$$
for all $t \in \mathbb{R}$.

Remark D.1. Objects in **Dyn** are smooth dynamical systems. Morphisms Ψ are *dynamic-preserving* (equivalently, *equivariant* or *trajectory-intertwining*). They are dynamic-preserving in the sense that for smooth manifolds and flows:

$$(\mathcal{M}, \varphi^t),\ (\mathcal{N}, \psi^t) \qquad\qquad t \in \mathbb{R} \tag{61}$$

the morphism,

$$\Psi = (\mathcal{M}, \varphi^t) \to (\mathcal{N}, \psi^t) \tag{62}$$

is a C^k $(k \geq 1)$ map that is equivariant,

$$\Psi(\varphi^t(x)) = \psi^t(\Psi(x)) \qquad\qquad \forall x \in \mathcal{M} \tag{63}$$

so that Ψ transports trajectories φ^t onto trajectories ψ^t:

$$\Psi \circ \varphi^t = \psi^t \circ \Psi \tag{64}$$

Example D.1. Take the two-torus with linear flow,

$$(\mathbb{T}^2, \varphi^t), \qquad\qquad \varphi^t(x, y) = (x + \alpha t,\ y + \beta t) \pmod 1 \tag{65}$$

and the circle with rigid rotation,

$$(\mathbb{S}^1, \psi^t), \qquad\qquad \psi^t(\theta) = \theta + \alpha t \pmod 1 \tag{66}$$

Let,

$$\Psi : \mathbb{T}^2 \to \mathbb{S}^1, \qquad\qquad \Psi(x, y) = x \bmod 1 \tag{67}$$

For all $(x, y) \in \mathbb{T}^2$ $t \in \mathbb{R}$:

$$\begin{aligned} \Psi(\varphi^t(x, y)) &= \Psi(x + \alpha t, y + \beta t) && (68)\\ &= (x + \alpha t) \pmod 1 && (69)\\ &= \psi^t(\Psi(x, y)) && (70) \end{aligned}$$

so,

$$\Psi \circ \varphi^t = \psi^t \circ \Phi, \qquad\qquad \forall t \in \mathbb{R} \tag{71}$$

and Ψ is a dynamic-preserving C^∞ map and morphism of **Dyn**.

Remark D.2 (Discrete–time systems via suspension). Although **Dyn** is defined only for *flows*, any discrete–time diffeomorphism $T\colon \mathcal{M} \to \mathcal{M}$ can be embedded in **Dyn** through its *suspension flow.*

Form the mapping torus

$$\mathrm{Sus}(\mathcal{M}, T) := (\mathcal{M} \times [0,1])/\big(x,1\big) \sim \big(Tx, 0\big), \tag{72}$$

and define the flow

$$\varphi^t[(x,u)] = [(T^{\lfloor t+u \rfloor}x,\ u + t - \lfloor t+u \rfloor)] \tag{73}$$

Then $(\mathrm{Sus}(\mathcal{M}, T), \varphi^t)$ is an object of **Dyn** whose *time*–1 *map* reproduces T:

$$\varphi^1[(x,0)] = [(Tx, 0)] \tag{74}$$

This construction extends to morphisms by $\Sigma(f)[(x,u)] = [(f(x), u)]$, producing a functor $\Sigma : \mathbf{Dyn}_{\mathbb{Z}} \to \mathbf{Dyn}$. As such, results for continuous–time objects apply *mutatis mutandis* to discrete systems via Σ.

Definition D.2 (**Vect**)**.** The category **Vect** consists of

- **objects** V finite-dimensional real vector spaces.
- **morphisms** $L : V \to W$ linear maps.

Definition D.3 (Embedding functor)**.** Fix, for every Dyn–object $(\mathcal{M}, \varphi^t)$ a generic embedding $\Phi_{\mathcal{M}} : \mathcal{M} \hookrightarrow \mathbb{R}^{m(\mathcal{M})}$. Define the functor

$$F : \mathbf{Dyn} \longrightarrow \mathbf{Vect} \quad \text{by} \quad \begin{cases} F\big(\mathcal{M}, \varphi^t\big) = \mathbb{R}^{m(\mathcal{M})}, \\ F(\Psi) = \Phi_{\mathcal{N}} \circ \Psi \circ \Phi_{\mathcal{M}}^{-1} : \mathbb{R}^{m(\mathcal{M})} \longrightarrow \mathbb{R}^{m(\mathcal{N})}. \end{cases}$$

with identity $F(\mathrm{id}_{(\mathcal{M},\varphi^t)}) = \mathrm{id}_{\mathbb{R}^{m(\mathcal{M})}}$.

Functoriality is immediate from $\Phi_{\mathcal{N}} \circ (\Psi_2 \circ \Psi_1) = F(\Psi_2) \circ F(\Psi_1)$, which gives the following diagram:

$$\begin{tikzcd}
(\mathcal{M}, \varphi^t) \arrow[r, "\Phi := \Phi_{\mathcal{M}}"] \arrow[d, "\Psi"'] & \mathbb{R}^{m(\mathcal{M})} \arrow[d, "\Phi' \circ \Psi \circ \Phi^{-1}"] \\
(\mathcal{N}, \psi^t) \arrow[r, "\Phi' := \Phi_{\mathcal{N}}"] & \mathbb{R}^{m(\mathcal{N})}
\end{tikzcd}$$

Definition D.4 (**Dyn** Component–wise product)**.** Let $\big(\mathcal{M}_i, \varphi_i^t\big)_{i=1}^n$ be objects of **Dyn**. Define

$$\mathcal{M} \ := \ \prod_{i=1}^n \mathcal{M}_i, \qquad \varphi^t(x_1, \ldots, x_n) \ := \ \big(\varphi_1^t(x_1), \ldots, \varphi_n^t(x_n)\big), \tag{75}$$

and projection maps $\pi_j(x_1, \ldots, x_n) = x_j \quad (j = 1, \ldots, n)$. We call

$$(\mathcal{M}, \varphi^t) \tag{76}$$

the *component–wise product* of the given objects.

Lemma D.1 (Universal property of finite products in **Dyn**)**.** *For the system $(\mathcal{M}, \varphi^t)$ and projections π_j defined above, the pair*

$$\big((\mathcal{M}, \varphi^t),\ \{\pi_j\}_{j=1}^n\big) \tag{77}$$

is the categorical product of $\big(\mathcal{M}_i, \varphi_i^t\big)_{i=1}^n$ in the category **Dyn**.

Proof. Because each π_j is smooth,

$$\pi_j\big(\varphi^t(x_1, \dots, x_n)\big) = \varphi_j^t(x_j) = \varphi_j^t\big(\pi_j(x_1, \dots, x_n)\big), \tag{78}$$

so π_j intertwines the flows and is therefore a morphism in **Dyn**.

Let $(\mathcal{N}, \psi^t)$ be any object and suppose we are given morphisms $f_j : (\mathcal{N}, \psi^t) \to (\mathcal{M}_j, \varphi_j^t)$ for $j = 1, \dots, n$. Define

$$f : \mathcal{N} \longrightarrow \mathcal{M}, \qquad f(y) := \big(f_1(y), \dots, f_n(y)\big). \tag{79}$$

The map f is C^k because each f_j is. Equivariance follows component–wise:

$$f(\psi^t y) = \big(f_1(\psi^t y), \dots\big) = \big(\varphi_1^t f_1(y), \dots\big) = \varphi^t\big(f(y)\big). \tag{80}$$

Hence f is a morphism in **Dyn** and $\pi_j \circ f = f_j$ by construction.

If $g : (\mathcal{N}, \psi^t) \to (\mathcal{M}, \varphi^t)$ is another morphism with the same property $\pi_j \circ g = f_j$ for every j, then for each $y \in \mathcal{N}$

$$g(y) = \big(\pi_1 g(y), \dots, \pi_n g(y)\big) = \big(f_1(y), \dots, f_n(y)\big) = f(y),$$

so $g = f$.

Since a unique mediating morphism exists for every candidate cone, $(\mathcal{M}, \varphi^t)$ with the projections π_j satisfies the universal property of finite products in **Dyn**. □

Lemma D.2 (Coproducts)**.** *For any family $\big(\mathcal{M}_i, \varphi_i^t\big)_{i \in I}$, the disjoint union*

$$\Big(\bigsqcup_{i \in I} \mathcal{M}_i,\ \varphi^t(x) = \varphi_i^t(x)\ \text{if}\ x \in \mathcal{M}_i\Big)$$

with the canonical injections $\iota_i : \mathcal{M}_i \hookrightarrow \bigsqcup_j \mathcal{M}_j$ is the coproduct in **Dyn**.

Proof. Disjoint unions of compact manifolds are compact manifolds, the flow is C^k on each component, and the injections are equivariant by construction. The universal property follows from the universal property of disjoint unions in smooth manifolds. □

Lemma D.3 (Regular equalisers)**.** *Let $f, g : (\mathcal{M}, \varphi^t) \to (\mathcal{N}, \psi^t)$ be morphisms. Assume f and g are transverse to the diagonal $\Delta_{\mathcal{N}} \subset \mathcal{N} \times \mathcal{N}$. Then*

$$E := \{x \in \mathcal{M} \mid f(x) = g(x)\}$$

is an embedded C^k sub-manifold, invariant under φ^t. The inclusion $i : E \hookrightarrow \mathcal{M}$ with the restricted flow $\varphi^t_{|E}$ realises the equaliser of (f, g) in **Dyn**.

Proof. Transversality implies E is an embedded sub-manifold. Equivariance of f and g gives $f(\varphi^t x) = \psi^t f(x) = \psi^t g(x) = g(\varphi^t x)$, so E is flow-invariant. The inclusion i is a morphism, and any morphism $h : (\mathcal{K}, \chi^t) \to (\mathcal{M}, \varphi^t)$ with $fh = gh$ factors uniquely through i by the usual smooth-manifold argument. □

Remark D.3 (Limitations of completeness). The category of C^k manifolds is neither complete nor cocomplete; an infinite product need not be a manifold, and general coequalisers (or non-regular equalisers) can fail to carry a smooth Hausdorff structure. Consequently **Dyn** is *finitely* complete and has coproducts, but it does not possess arbitrary small limits or colimits.

Lemma D.4 (Limits and colimits)**.** **Dyn** *has all small limits and colimits; they are inherited point-wise from the category of C^k manifolds.*

Sketch. Products exist because finite products of compact manifolds are compact manifolds; the flow acts component-wise. Equalisers are submanifolds defined by equalising maps that commute with the flow. Dually for colimits. □

The remainder of this section builds functors *out of* **Dyn** and uses natural transformations and comonads to formalise local linearisation and its residual correction.

D.2 Local linearisation as a natural transformation

Write $\Pi_d : \mathbf{Dyn} \to \mathbf{Dyn}$ for the functor that forgets all ≥ 2-jet data. Weighted least squares at each anchor x induces the component

$$\eta_{(\mathcal{M},\varphi^t)}(x) \;:=\; B_x : F(\mathcal{M},\varphi^t) \;\longrightarrow\; F \circ \Pi_d(\mathcal{M},\varphi^t),$$

yielding a natural transformation $\eta : F \Rightarrow F \circ \Pi_d$.

D.3 Residual frame field as a cofree coalgebra

Define a comonad $\mathsf{G} : (\mathbf{Vect}, \otimes) \to (\mathbf{Vect}, \otimes)$ by $\mathsf{G}(V) = V \oplus \big(\operatorname{span} Q_x\big)$. The map $x \mapsto (B_x, Q_x)$ endows each fibre with the structure of a G-coalgebra, cofree under the variance-fraction constraint ρ. In string-diagram notation:

$$\begin{tikzcd}
F(\mathcal{M},\varphi^t) \arrow[r, dashed, "\Delta"] \arrow[d, "\eta"'] & \mathsf{G}F(\mathcal{M},\varphi^t) \arrow[d, "\mathsf{G}\eta"] \\
F \circ \Pi_d(\mathcal{M},\varphi^t) \arrow[r, dashed, "\mathsf{G}\Delta"'] & \mathsf{G}F \circ \Pi_d(\mathcal{M},\varphi^t)
\end{tikzcd}$$

The dashed arrows are the Rose augmentation; commutativity is enforced by letting G be *cofree* with respect to ρ.